EXCEL TABLES + EXERCISES

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Introduction to Excel Tables

- An Excel table is a set of data organized in the form of rows and columns and converted into a "structured table" using the Insert > Table command (Ctrl + T).
- An Excel table has its own context menu.
- In short, an Excel table is not just an area of cells: it is an intelligent object that facilitates the management, analysis and presentation of data.

Benefits of Excel Tables

- Automatic and consistent formatting
  - Consistent style (alternating bands, frozen headers).
  - Advantage: time saving and better readability.

- Dynamic Data Management
  - When the size of the data changes regularly (adding/removing rows).
  - Advantage: Formulas, charts, and pivot tables update automatically.

- Structured References in Formulas
  - No need for classic references (A1:B10), use column names.
  - Advantage: more readable and robust formulas.

Benefits of Excel Tables

- Need for quick filters and sorting
  - If you need to filter or sort your data frequently without creating manual filters.
  - Advantage: Built-in filters and sorting to organize your data.

- Working with slicers
  - If you want to visually filter your data.
  - Advantage: segments available only for tables and TCDs.

- Creating Automatic Calculated Columns
  - Add calculations on each row without manually copying the formula.
  - Advantage: A single formula applies to the entire column.

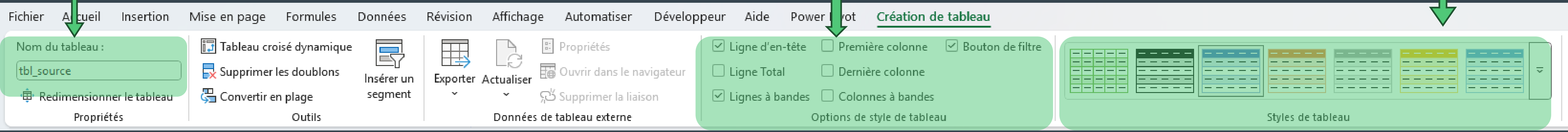
Benefits of Excel Tables

- Data preparation for Pivot Tables
  - When your data is intended to be analyzed via TCD.
  - Advantage: DBTs are automatically updated with new data.

- Collaboration and automation
  - When the file is shared or automated (Power Query, Power BI).
  - Advantage: BI tools recognize tables as reliable sources.

Exercise 1: Range -> Table

- Duplicate the cell range of the 'Source' sheet into the 'Ex1' sheet and turn it into a structured table. Note: the automatic formatting, the context menu and the sorting and filter menu in each column.
- Rename it 'tbl_source'.
- Scroll down in the table and notice that the headings remain visible in the column index bar.
- Change the formatting by clicking on one of the suggested styles.
- In the 'Table Creation' menu, explore the Style Options group.



Exercise 1: Table, filters and sorting

- Sort the table by [Product Name] in alphabetical order.
- Sort by Number: Sort the table by [Unit Price] (ascending) then by [Nb in stock] (descending). Note that simple sorting is exclusive.
- Multi-criteria sorting: Sort by [CategoryID] (ascending) and then by [Nb in stock] (ascending). You need to go through Sort by Color > Custom Sort.
- Filter the table to show only products in the 'Drinks' category.
- Filter [Packaging] where the value starts with 24.
- Filter [Packaging] where the value contains 'kg'.
- Filter [1st sale] on Q2 of 2025.
- Apply conditional formatting to [No. in stock] with a set of icons, and then filter by color to show only the lowest values.

Exercise 1: Table, filters and sorting

- Apply conditional formatting to [ProviderID] and filter on a color from the available selection.
- Clear the current filters and insert a text, a text box and an image (from your website for example). Filter the table on the 'Dessert' category AND on the supplier 11. What do you see?
- Question: You are asked to filter the table to show the 10 lowest items in stock that have a price under 20.
- Question: Filter the table to show only the values in [packaging] that contain the words 'l', 'cl' or 'ml'. Use a custom filter with the double condition: either contains '?l?' or ends with '?l'.
- Apply custom sorting on [CategoryID] ascending and [Nb in stock] descending.

Exercise 2: Formulas in a table

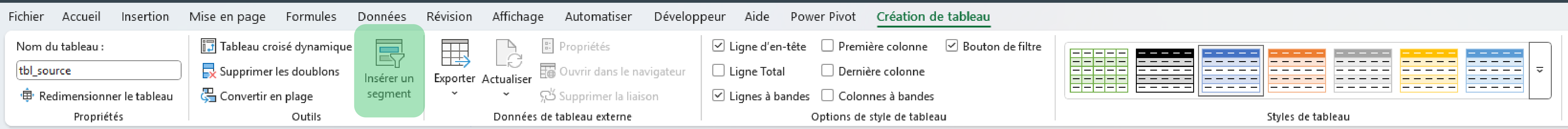
- In the 'Ex2 Formulas' sheet, duplicate the existing range and turn it into a table. Choose a custom style from the styles available in the 'Create Table' menu.
- In the range and table, add the necessary formulas in the TOTAL and TOTAL columns including VAT. The VAT rate is available on the 'Variables' sheet or under the defined name 'VAT rate'. Compare the two types of formula.
- In the Q6:R10 box, add the formulas needed to calculate the sum of the TOTAL and TOTAL TTC column as well as the total number of stores (For Range and Table).
- Add the new data (A33:D39) below the range and table. What do you see in the totals formulas?

Exercise 3: Table and chart

- In the 'Ex3 Chart' sheet, duplicate the existing range in H6 and turn it into a table.
- Insert two column charts that show the total sales by store, the 1st based on the cell range, the 2nd on the table.
- Add the new data to the bottom of the table and range and compare the impact on the two charts.
- Remove rows from the table by right-clicking on one or more rows.

Exercise 4: Slicers

- In the 'Segment' sheet, copy the range from the 'Source' sheet and turn it into a table.
- Insert three segments by selecting [CategoryID], [VendorID], and [Sale Completed].
- Format the segments using the associated context menu.
- Select different values in the segments to filter the table.
- Apply conditional formatting that highlights [Nb in stock] values that are less than 5.

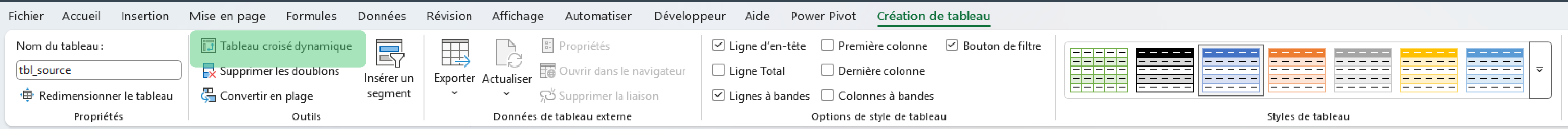


Exercise 5: Drop-down list

- In the 'Drop-down list' sheet, copy the [CategoryID] column from the 'Source' sheet and turn it into a table.
- Use the 'Remove Duplicates' function in this table.
- Rename it tbl_cat.
- Create a drop-down list (Data > Data Validation > List) that points to the [CategoryID] column.
- Create a defined name 'Liste_Cat' on the [CategoryID] column and add another drop-down list that points to that defined name.
- Add values to the table and check for updated drop-down lists.
- In comparison, copy the [CategoryID] column and convert it to a range from which you will create a new drop-down list. Add values and compare the update of the drop-down lists.

Exercise 6: Pivot Table

- Click in a cell in the table in Exercise 1 and click Table Creation > Pivot Table. Create the Pivot table on the 'Ex6 TCD' sheet in A1 for example.

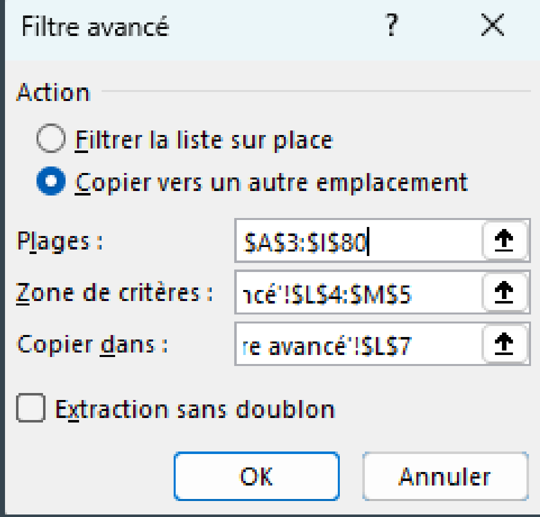


- In the Pivot table, display the sum of [Nb in Stock] (Values) broken down by [CategoryID] (Rows).
- Insert a bar chart from the Pivot table.
- Insert a slicer on [ProviderID] and a timeline (date segment) on [1st Sale].
- Apply conditional formatting that highlights calculated values that are higher than the average.

- REMINDER: Put your data in an Excel table if you are planning an analysis or import of this data through a TCD, Power Query, Power Pivot, or Power BI.

Exercise 7: Advanced AND Filter

1. AND Filter: Create a criteria area with the column headers you want to filter. For example, [ProviderID] and [CategoryID]. Enter filter values below the headers (e.g. supplierID=2 and CategoryID="Spices").
2. Then apply an advanced filter: Data > (Sort and Filter) Advanced. Select the option 'Copy to another location' and then in 'Ranges', select the table, in 'Criteria Area', select the cells containing the headers and filter values and in 'Copy to', select the cell where you want to see the result.



1/ Filtre ET								
FournisseurID	CategoryID							
2	Epices							
ProductID	Nom Produit	FournisseurID	CategoryID	Packaging	Prix Unit.	Nb en sto	Vente ter	1ère vente
4	Chef Anton's Cajun Seasoning	2	Epices	48 - 6 oz ja	22	53	FAUX	02.01.2025
5	Chef Anton's Gumbo Mix	2	Epices	36 boxes	21.35	0	VRAI	11.12.2025
65	Louisiana Fiery Hot Pepper Sauce	2	Epices	32 - 8 oz b	21.05	76	FAUX	27.02.2025
66	Louisiana Hot Spiced Okra	2	Epices	24 - 8 oz ja	17	4	FAUX	17.12.2025

Exercise 7: Advanced Filter OR

1. Filter OR: Copy the column headers and enter the same values as before but on two separate rows. Then apply an advanced filter which will filter the rows that meet condition 1 OR condition 2.

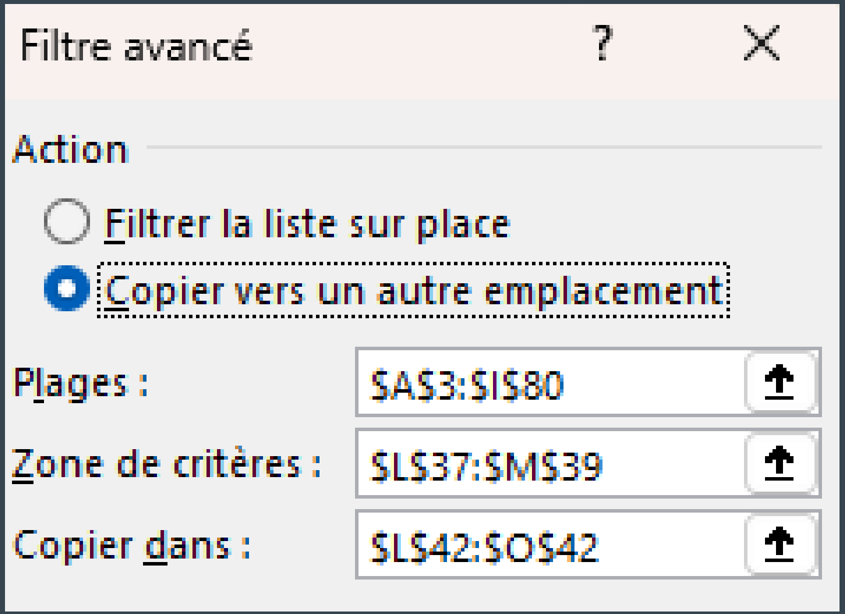
FournisseurID	CategoryID
2	
	Epices

2/ Filtre OU

FournisseurID	CategoryID
2	
	Epices

ProductID	Nom Produit	FournisseurID	CategoryID	Packaging	Prix Unit.	Nb en sto	Vente ter	1ère vente
3	Aniseed Syrup	1	Epices	12 - 550 m	10	13	FAUX	05.04.2025
4	Chef Anton's Cajun Seasoning	2	Epices	48 - 6 oz ja	22	53	FAUX	02.01.2025
5	Chef Anton's Gumbo Mix	2	Epices	36 boxes	21.35	0	VRAI	11.12.2025
6	Grandma's Boysenberry Spread	3	Epices	12 - 8 oz ja	25	120	FAUX	31.07.2025
8	Northwoods Cranberry Sauce	3	Epices	12 - 12 oz j	40	6	FAUX	19.07.2025
15	Genen Shouyu	6	Epices	24 - 250 m	15.5	39	FAUX	10.09.2025
44	Gula Malacca	20	Epices	20 - 2 kg b	19.45	27	FAUX	19.12.2025
61	Sirop d'érable	29	Epices	24 - 500 m	28.5	113	FAUX	06.11.2025
63	Vegie-spread	7	Epices	15 - 625 g j	43.9	24	FAUX	16.06.2025
65	Louisiana Fiery Hot Pepper Sauce	2	Epices	32 - 8 oz b	21.05	76	FAUX	27.02.2025
66	Louisiana Hot Spiced Okra	2	Epices	24 - 8 oz ja	17	4	FAUX	17.12.2025
77	Original Frankfurter grüne Soße	12	Epices	12 boxes	13	32	FAUX	14.08.2025

Exercise 7: Advanced filter AND+ OR



1/ Filtre ET + OU			
FournisseurID	CategoryID		
>10	Epices		
	Fromage		
Résultat			
Nom Produit	Prix Unit.	Nb en stock	1ère vente
Queso Cabrales	21	22	15.11.2025
Queso Manchego	38	86	25.08.2025
Gorgonzola Telin	12.5	0	14.05.2025
Mascarpone Fabi	32	9	20.12.2025
Geitost	2.5	112	27.05.2025
Gula Malacca	19.45	27	19.12.2025
Raclette Courdav	55	79	14.11.2025
Camembert Pier	34	19	24.04.2025
Sirop d'érable	28.5	113	06.11.2025
Gudbrandsdalsos	36	26	19.06.2025
Flotemysost	21.5	26	13.12.2025
Mozzarella di Gio	34.8	14	01.03.2025
Original Frankfur	13	32	14.08.2025

1. Filter AND+OR: Copy the criteria area with the values opposite. Apply the advanced filter to see the filtered rows corresponding to the "Spices" category where [supplierID] is > 10 OR [CategoryID] is "Cheese".
2. Copy the headers of the columns you want to display in the result below the criteria area.
3. Select the column headers in the 'Copy to:' area of the Advanced Filter dialog

Exercise 8: Filter() + Sort() Functions

1. Use the FILTER() matrix function to view the filtered rows from the original table in this sheet where the [Number in Stock] column is less than or equal to 3.

2. Use the sort matrix function to sort the original table against the [1st sale] column.

3. Use a combination of the two functions to display the sorted table at point 2 where the column [No. in stock] is <= to 3.

- =FILTER(range; include; [if_empty])
- SORT(range; [sort_index]; [order]; [by_column])